

LAURENS COUNTY AP, SC, USA

WMO: 720597

Lat: 34.500N

Lon: 81.950W

Elev: 212

StdP: 98.80

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00190

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.3	(c) -2.7	(d) -14.8	(e) 1.1	(f) -2.0	(g) -12.1	(h) 1.4	(i) 0.1	(j) 8.8	(k) 14.0	(l) 8.0	(m) 12.3	(n) 1.8	(o) 20	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.9	24.5	32.9	23.8	32.4	23.6	27.4	30.8	26.3	29.4	25.3	29.1	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.3	29.2	25.2	20.9	27.9	24.2	19.6	26.9	87.7	31.4	83.4	29.6	78.0	28.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.0	(c) 5.2	(e) -9.6	(f) 36.1	(g) 4.8	(h) 1.2	(i) -13.1	(j) 37.0	(k) -15.9	(l) 37.7	(m) -18.6	(n) 38.3	(o) -22.1	(p) 39.2		
(a) 7.4	(b) 6.0	(c) 5.2	(e) -10.8	(f) 26.9	(g) 4.4	(h) 1.3	(i) -13.9	(j) 27.8	(k) -16.5	(l) 28.6	(m) -19.0	(n) 29.4	(o) -22.2	(p) 30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	6.0	8.4	11.6	16.7	21.1	25.1	26.4	25.6	23.1	16.7	10.8	8.0		
	DBStd	8.23	5.21	5.23	5.01	3.97	3.44	2.38	1.97	2.01	2.97	4.21	4.07	4.93		
	HDD10.0	418	146	86	42	3	0	0	0	0	0	3	38	101		
	HDD18.3	1606	384	279	215	78	17	0	0	0	3	80	227	322		
	CDD10.0	2851	21	41	91	203	346	452	509	485	392	211	62	39		
	CDD18.3	996	0	2	6	28	104	202	251	227	145	30	1	1		
	CDH23.3	8861	1	7	60	275	862	1864	2399	1965	1171	250	7	1		
	CDH26.7	3576	0	0	5	47	271	797	1101	834	462	59	0	0		
(14) Wind	WSAvg	2.2	2.4	2.7	2.7	2.7	2.2	1.9	1.9	1.6	1.7	2.1	2.0	2.2		
(15) Precipitation	PrecAvg	1209	117	107	131	92	92	104	90	104	96	87	90	97		
	PrecMax	1716	198	205	277	266	258	322	260	274	233	360	263	222		
	PrecMin	824	15	14	24	16	14	7	15	2	0	0	12	18		
	PrecStd	184	44	45	59	53	54	64	52	63	57	72	50	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	24.1	27.4	30.0	33.0	35.2	36.2	35.2	34.0	32.0	24.0	22.4		
		MCWB	16.4	17.1	16.8	19.4	21.2	25.4	25.2	24.1	23.0	22.8	17.9	17.7		
	2%	DB	18.0	22.0	25.2	27.8	31.3	33.9	35.0	33.0	32.5	27.7	22.2	20.3		
		MCWB	14.4	16.1	16.1	18.3	21.1	24.3	24.9	23.7	22.7	20.2	16.0	17.5		
	5%	DB	17.1	19.8	22.6	27.0	29.9	32.5	33.0	32.4	31.3	26.1	20.2	17.9		
		MCWB	13.7	14.3	15.0	18.2	21.3	23.8	24.5	23.6	22.4	19.2	15.4	15.7		
	10%	DB	14.8	17.3	21.0	24.9	27.7	31.2	32.3	31.2	29.9	23.9	17.9	16.4		
		MCWB	11.1	13.3	14.3	17.1	20.8	23.4	24.4	23.5	22.1	18.6	14.2	13.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	18.7	19.0	22.3	24.7	28.0	28.4	28.4	25.5	23.9	20.0	20.3		
		MCDB	19.1	22.6	23.4	26.5	28.5	33.2	33.4	31.8	29.2	27.5	21.1	21.0		
	2%	WB	16.5	17.5	17.8	20.8	23.5	27.1	27.3	27.2	24.6	22.7	18.2	18.0		
		MCDB	18.1	19.5	21.7	24.0	27.7	31.5	30.9	29.0	29.2	26.1	20.5	19.2		
	5%	WB	14.2	15.8	16.9	19.9	22.5	25.9	26.2	25.7	23.6	21.5	17.2	16.3		
		MCDB	16.6	18.5	20.7	24.0	26.8	29.4	29.7	28.4	28.4	24.0	18.5	17.3		
	10%	WB	11.8	13.8	15.7	18.8	21.9	24.5	25.2	24.8	22.9	19.9	15.5	14.1		
		MCDB	13.6	16.6	19.2	23.3	26.0	28.5	29.7	28.7	27.5	22.9	17.4	16.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.9	12.2	12.9	11.5	11.1	10.9	10.5	11.1	12.2	11.9	9.8		
		MCDBR	12.7	13.0	14.3	14.0	13.6	12.7	12.4	12.2	13.6	13.5	13.8	10.5		
	5% WB	MCWBR	8.8	7.9	7.2	6.1	5.4	4.7	4.1	4.2	5.0	6.7	8.5	7.5		
		MCWBR	10.6	10.7	12.0	11.5	11.0	11.4	11.4	10.0	11.7	10.8	10.0	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.325	0.361	0.401	0.459	0.487	0.535	0.520	0.429	0.390	0.341	0.315		
	taud		2.501	2.474	2.399	2.279	2.158	2.121	2.011	2.040	2.274	2.360	2.457	2.532		
	Ebn at Noon		895	924	911	886	835	807	766	770	831	837	854	874		
	Edh at Noon		84	96	112	133	152	157	175	167	126	106	87	77		
(44) All-Sky Solar Radiation	RadAvg		2.70	3.33	4.42	5.53	6.12	6.46	6.07	5.56	4.77	3.96	2.99	2.33		
	RadStd		0.14	0.38	0.31	0.41	0.49	0.46	0.39	0.27	0.42	0.47	0.26	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+117

Nomenclature: See separate page